Grip or glide? The surface challenge · Teacher and Cover Plan

Learning goal: I can explain how friction acts between touching surfaces and judge when grip or glide is useful.

Scheme Outcome

TRACK Science scheme of work: Forces and Friction

Explain friction as a contact force opposing relative sliding and evaluate when friction is helpful or unhelpful.

Success Criteria

I can identify the two touching surfaces where friction acts.

I can draw a friction arrow opposite the direction of sliding.

I can explain one job where friction helps and one where it is unhelpful.

Model Sequence (I do)

Friction is a contact force between two surfaces that slide, or try to slide, past each other.

The force acts opposite to the relative sliding and depends on both materials in the surface pair.

Grip is helpful for brakes and shoe soles, whereas unwanted rubbing heats and wears machine parts.

On a moving box sliding right, draw the friction arrow

pointing left along the contact surface.

On a shoe pushing backwards against the ground, grip prevents slipping so you can walk forwards.

Never label friction simply 'good' or 'bad'·always name the two surfaces and the job.

We Do & Hinge Check

Which statement about friction is scientifically accurate?

Answer: A. Smooth surfaces still experience friction; whether it helps depends on the surface pair and the job.

Safe Practical & Card Setup

Examine the surface cards or safe sample strips. Draw a force diagram and explain one helpful and one unhelpful example.

Kit: Safe texture samples on a tray (rubber sole, smooth plastic, carpet strip, card) or surface scenario cards.

Inspect samples on the desk tray. Do not rub skin hard against rough surfaces or slides.

You Do Routes (Supported, Independent, Aspire)

Supported: Add a friction arrow opposite the motion arrow. Finish one sentence explaining how brake pads help. (Help: The box slides right, so friction points __. Brake pads grip the wheel to help it __.)

Independent: Draw a box sliding right with a friction

arrow. Explain one helpful and one unhelpful friction example using the two touching surfaces. (Help: Friction acts between __ and __. It opposes sliding, which helps / hinders because __.)

Aspire: Explain how static grip helps a person walk forwards while kinetic friction slows a sliding box. (Help: Friction opposes relative slipping at the contact: when a foot pushes backwards on the ground, grip prevents backward slip.)